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Experiment:07

AIM:Implement and analyze various probability distributions in Python, including Binomial, Poisson, and Normal distributions.

THEORY:

1. Difference between Binomial, Poisson, and Normal distributions? **ANS:**Binomial distribution is a discrete distribution that represents the number of successes in a fixed number of independent trials, where each trial has two outcomes (success or failure) and constant probability. Poisson distribution is also discrete but is used to model the number of events occurring in a fixed interval of time or space with a known average rate and independent occurrences. Normal distribution is a continuous distribution that is symmetric and bell-shaped, defined by its mean and standard deviation, and is used to represent natural variations in data.
2. What is Real-world significance of Poisson distribution? **ANS:**Poisson distribution is important for modeling the occurrence of random events over a given interval of time or space. It is widely used in real-life situations such as counting the number of customers arriving at a store, number of phone calls received in a call center, or number of defects in a product. It helps organizations in planning, forecasting, and decision-making, especially in fields like telecommunications, traffic management, and reliability engineering.
3. What is Central Limit Theorem (CLT) and how does it relate to Normal Distribution? **ANS:**The Central Limit Theorem states that when a large number of independent samples are taken from any population, the distribution of the sample means approaches a normal distribution, regardless of the original population distribution. This theorem is important because it allows statisticians to apply normal distribution methods to a wide range of problems, even when the data is not normally distributed, as long as the sample size is sufficiently large.
4. Why Normal distribution is widely used in machine learning and statistics? **ANS:**Normal distribution is widely used because many real-world phenomena, such as heights, weights, and measurement errors, naturally follow a normal pattern. It is mathematically simple and has well-defined properties, making it easy to analyze and interpret. Additionally, many statistical techniques and machine learning algorithms assume normality, and the Central Limit Theorem further supports its use in practical applications.
5. What is Skewness in distributions and which distributions exhibit it? **ANS:**Skewness is a measure of the asymmetry of a probability distribution. A distribution is positively skewed if it has a long tail on the right side and negatively skewed if the

tail is on the left side. In skewed distributions, the mean, median, and mode are not equal. Binomial and Poisson distributions can show skewness depending on their parameters, while the Normal distribution is perfectly symmetric and has zero skewness.

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import binom, poisson, norm

# Set up a figure with subplots
fig, axs = plt.subplots(3, 3, figsize=(15, 12))
fig.suptitle("Analysis of Binomial, Poisson, and Normal
Distributions", fontsize=16)

# ----- BINOMIAL DISTRIBUTION -----
n, p = 10, 0.5
x_binom = np.arange(0, n+1)
pmf_binom = binom.pmf(x_binom, n, p)
cdf_binom = binom.cdf(x_binom, n, p)
samples_binom = binom.rvs(n, p, size=1000)

# PMF
axs[0, 0].bar(x_binom, pmf_binom, color='skyblue')
axs[0, 0].set_title("Binomial PMF (n=10, p=0.5)")
# CDF
axs[0, 1].step(x_binom, cdf_binom, where='mid', color='blue')
axs[0, 1].set_title("Binomial CDF")
# Histogram of Samples
axs[0, 2].hist(samples_binom, bins=np.arange(n+2)-0.5, density=True,
color='navy', alpha=0.7)
axs[0, 2].set_title("Binomial Samples Histogram")

# ----- POISSON DISTRIBUTION -----
mu = 4
x_poisson = np.arange(0, 15)
pmf_poisson = poisson.pmf(x_poisson, mu)
cdf_poisson = poisson.cdf(x_poisson, mu)
samples_poisson = poisson.rvs(mu, size=1000)

# PMF
axs[1, 0].bar(x_poisson, pmf_poisson, color='lightgreen')
axs[1, 0].set_title("Poisson PMF ( $\mu=4$ )")
# CDF
axs[1, 1].step(x_poisson, cdf_poisson, where='mid', color='green')
axs[1, 1].set_title("Poisson CDF")
# Histogram of Samples
axs[1, 2].hist(samples_poisson, bins=np.arange(16)-0.5, density=True,
color='darkgreen', alpha=0.7)
axs[1, 2].set_title("Poisson Samples Histogram")
```

```

# ----- NORMAL DISTRIBUTION -----
mu_norm, sigma = 0, 1
x_norm = np.linspace(-4, 4, 1000)
pdf_norm = norm.pdf(x_norm, mu_norm, sigma)
cdf_norm = norm.cdf(x_norm, mu_norm, sigma)
samples_norm = norm.rvs(mu_norm, sigma, size=1000)

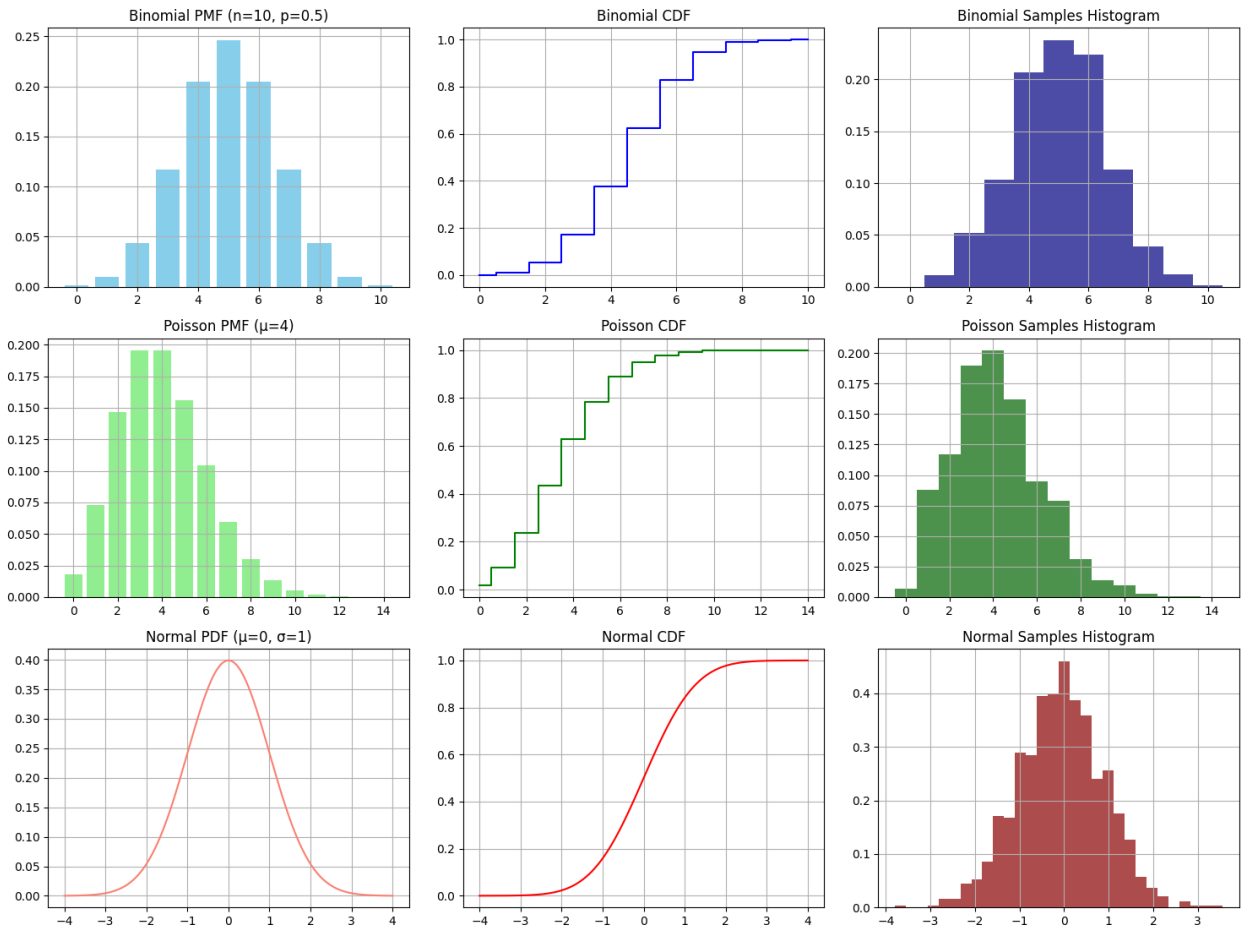
# PDF
axs[2, 0].plot(x_norm, pdf_norm, color='salmon')
axs[2, 0].set_title("Normal PDF ( $\mu=0$ ,  $\sigma=1$ )")
# CDF
axs[2, 1].plot(x_norm, cdf_norm, color='red')
axs[2, 1].set_title("Normal CDF")
# Histogram of Samples
axs[2, 2].hist(samples_norm, bins=30, density=True, color='darkred',
alpha=0.7)
axs[2, 2].set_title("Normal Samples Histogram")

# Layout and show
for row in axs:
    for ax in row:
        ax.grid(True)
plt.tight_layout(rect=[0, 0, 1, 0.96])
plt.show()

# ----- BASIC STATISTICAL ANALYSIS -----
print("Binomial Distribution:")
print(f"Mean = {np.mean(samples_binom):.2f}, Variance = {np.var(samples_binom):.2f}")
print("\nPoisson Distribution:")
print(f"Mean = {np.mean(samples_poisson):.2f}, Variance = {np.var(samples_poisson):.2f}")
print("\Normal Distribution:")
print(f"Mean = {np.mean(samples_norm):.2f}, Standard Deviation = {np.std(samples_norm):.2f}")

```

Analysis of Binomial, Poisson, and Normal Distributions



Binomial Distribution:
Mean = 5.01, Variance = 2.51

Poisson Distribution:
Mean = 4.17, Variance = 4.29

Normal Distribution:
Mean = -0.08, Standard Deviation = 0.96

Conclusion: Probability distributions such as Binomial, Poisson, and Normal play an important role in statistics and data analysis by helping us understand and model different types of data. The Binomial and Poisson distributions deal with discrete events, while the Normal distribution describes continuous data and natural variations. Concepts like the Central Limit Theorem explain why the Normal distribution is widely applicable. Additionally, measures like skewness help in understanding the shape of data. Overall, these distributions are essential tools for analyzing real-world problems and making informed decisions in fields like science, engineering, and machine learning.